MILITARY STANDARD

TRUCKLOADING OF HAZARDOUS MATERIALS

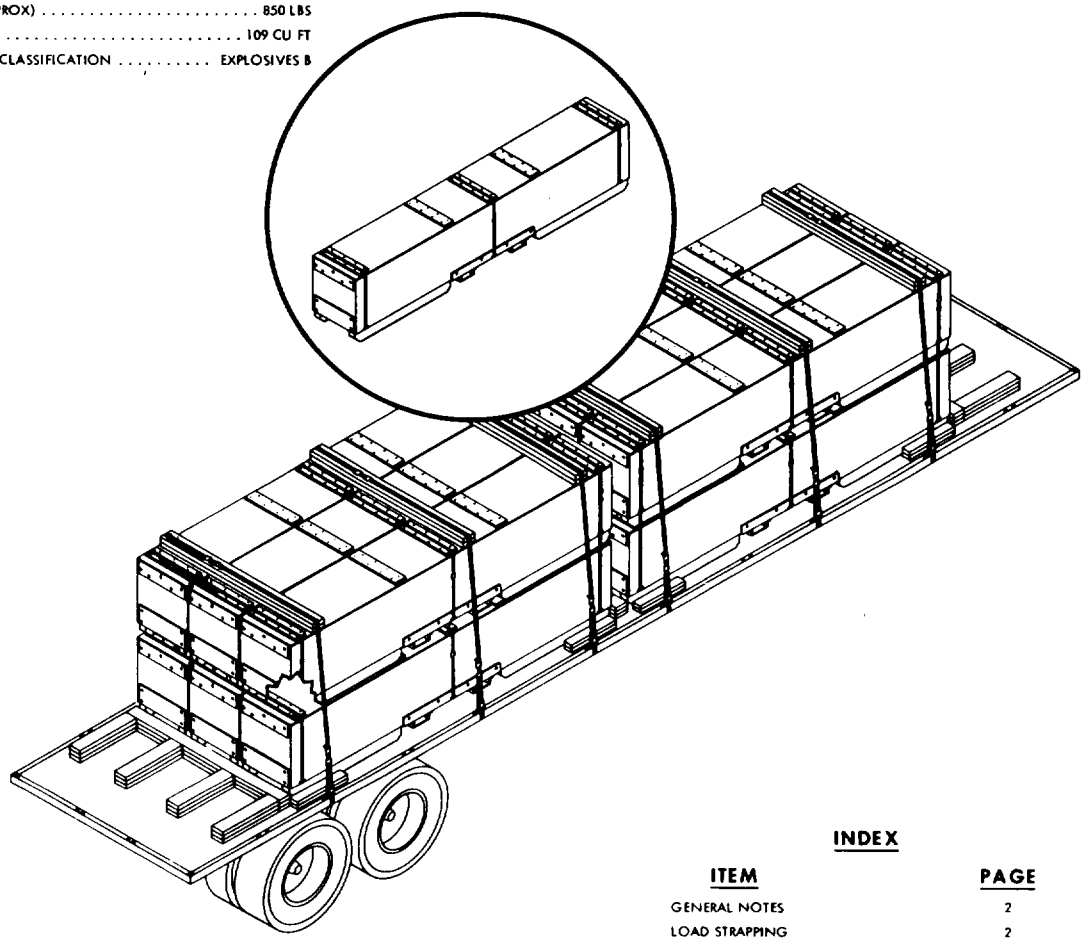
BOOSTER, MK 12 MOD 0 & 1
IN CONTAINER MK 578 MOD 0

MIL-STD-1320-103
(NAVY)

9 JUNE 1975

CONTAINER DATA

DIMENSIONS . . . . . 175 L x 28 W x 38 1/2 H
GROSS WT. (APPROX) . . . . . 2690 LBS
TARE WT. (APPROX) . . . . . 850 LBS
CUBE . . . . . 109 CU FT
DOT HAZARD CLASSIFICATION . . . . . EXPLOSIVES B



NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINER-LOADING AND CARLOADING MILITARY STANDARDS. REFER TO INDEX TO STANDARDS, MIL-HDBK-236 (NAVY).

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FSC 8140

AUTHORIZED AND RELEASED FOR HIGHWAY SHIPMENTS ONLY

M.S. Gray NWHG 6/2/75
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

J.E. Kelly 6/2/75
SIGNATURE NAVSEASYSCOM BY DIRECTION DATE

ORIGINATOR [signature] 2.1.111
SIGNATURE

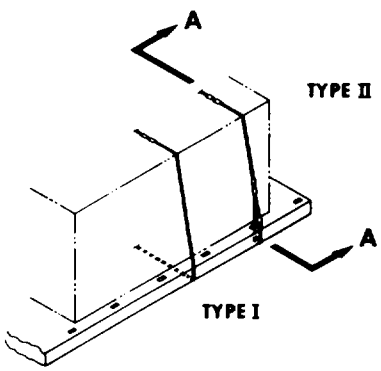
NAVAL WEAPONS HANDLING LABORATORY
N A D EARLE, NEW JERSEY

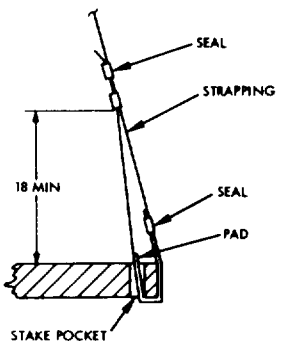
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GENERAL NOTES

1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING THE BOOSTER MK 12 MODS 0 AND 1 IN CONTAINER MK 578 MOD 0.
2. THE PROCEDURES AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 40 FT FLATBED TRAILERS WITH THE AXLES LOCATED IN THE "WESTERN" POSITION (AT THE EXTREME REAR OF THE TRAILER). DO NOT USE TRAILERS WITH METAL FLOORS.
3. CHAINS AND STEEL STRAPS MAY BE USED INTERCHANGEABLY FOR TIE-DOWNS ON A 1 TO 1 BASIS.
4. STEEL STRAPS SHALL BE 2 x .050 AND APPLIED AS SPECIFIED BY THE LOAD STRAPPING REQUIREMENTS SHOWN ON PAGE 2. CHAINS AND LOAD BINDERS SHALL BE APPLIED AS SHOWN ON PAGE 5.
5. WHEN USING CHAIN OR STRAP, INSERT A PROTECTOR BOARD (LAMINATED DOUBLE 2 x 6) BETWEEN CHAIN AND LADING. SECURE CHAIN TO THE BOARD WITH FOUR 20d NAILS BY DRIVING THE NAIL THROUGH THE CHAIN INTO THE BOARD AND CLINCHING IT OVER THE CHAIN.
6. CHAINS AND LOAD BINDERS SHALL MEET THE FOLLOWING CONDITIONS:
   a. CHAINS AND FITTINGS SHALL BE 3/8 INCH AND LOADBINDERS 3/8 INCH CAPACITY.
   b. CHAINS, FITTINGS AND LOADBINDERS SHALL HAVE A MINIMUM SAFE WORKING LOAD OF 5000 LBS.
   c. LOADBINDERS SHALL BE SAFETY WIRED WITH 16 GAUGE SOFT ANNEALED IRON WIRE OR SECURED USING THE END PIECE OF THE 3/8 CHAIN.
7. PRIOR TO LOADING THE TRAILER AND DURING THE PRELOADING INSPECTION REQUIRED BY OP 2165 AND REPORTED ON DD FORM 626, THE CHAINS, FITTINGS AND LOADBINDERS SHALL BE INSPECTED FOR STRETCH, GOUGING, BENT LINKS, WEAR AND ANY OTHER NOTICEABLE DEFECTS. THE INSPECTOR SHALL CONFIRM THAT THE CHAINS AND LOADBINDERS HAVE BEEN INSPECTED AND SHALL SO NOTE IN ITEM NO. 22 OF DD FORM 626 ANY DEFICIENCY SHALL BE CAUSE FOR REJECTION OF A CHAIN OR LOADBINDER.
8. THE LADING SHALL BE COVERED WITH A WATERPROOF, FIRE RESISTANT TARPAULIN. THE BEST RESULTS ARE OBTAINED IF THE CONTAINERS ARE COVERED WITH THE TARPAULIN PRIOR TO APPLYING THE TIE-DOWN CHAINS OR STRAPS. PROTECTOR BOARDS SHALL BE USED BETWEEN THE TIE-DOWNS AND TARPAULINS.
9. WHEN CONTAINERS ARE STACKED TWO HIGH, THEY SHALL BE UNITIZED AS SHOWN ON PAGE 4.
10. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS.
11. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS TO TRAILER IN AN ACCESSIBLE AREA AND ATTACH "EXPLOSIVES B" PLACARD TO BOTH SIDES, FRONT AND BACK OF TRAILER.
12. APPLICABLE MATERIAL SPECIFICATIONS: DUNNAGE LUMBER, MM-L-751; NAILS, FF-N-105, TYPE II STYLE 10, COMMON BRIGHT; STRAPPING, QQ-S-781, TYPE I CLASS A.
13. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).

LOAD STRAPPING





SECTION A-A

TIE-DOWN STRAPPING

THE FOLLOWING TWO METHODS OF TIE-DOWN STRAPPING ARE APPROVED FOR USE WITH THE SHIPMENT OF LOADS ON FLATBED TRAILERS. STEEL STRAPPING SHALL BE IN ACCORDANCE WITH FEDERAL SPECIFICATION QQ-S-781. STRAPPING SEALS SHALL HAVE A MINIMUM TENSILE STRENGTH EQUAL TO 75 PERCENT OF THE STRAP STRENGTH. FOR EACH OF THE TYPES ILLUSTRATED ABOVE IT IS PREFERRED TO POSITION, TENSION AND DOUBLE CRIMP THE STRAP SEALS AT THE TOP OF THE LOAD, IF PRACTICABLE.

TYPE I - CONTINUOUS STRAPPING AROUND THE FLATBED AND THE LOAD.

TYPE II - THE STRAPPING IS SECURED TO THE STAKE POCKETS, ONE PIECE ON EACH SIDE OF THE TRAILER, AND IS BROUGHT UP OVER THE LOAD, TENSIONED, AND SEALED WITH TWO DOUBLE-CRIMPED SEALS ON THE TOP. METHOD OF SECURING STRAPPING TO STAKE POCKET IS SHOWN IN SECTION A-A. THE SHORT END IS ON THE INSIDE AND IS SECURED WITH TWO DOUBLE-CRIMPED SEALS AT A MINIMUM OF 18 INCHES ABOVE THE TRAILER BED. A STAKE POCKET PAD (A SHORT PIECE OF THE SAME STRAPPING 18 INCHES LONG) IS INSERTED BETWEEN THE MAIN STRAP AND THE STAKE POCKET AND IS SECURED TO THE MAIN STRAP WITH A SEAL AS SHOWN.

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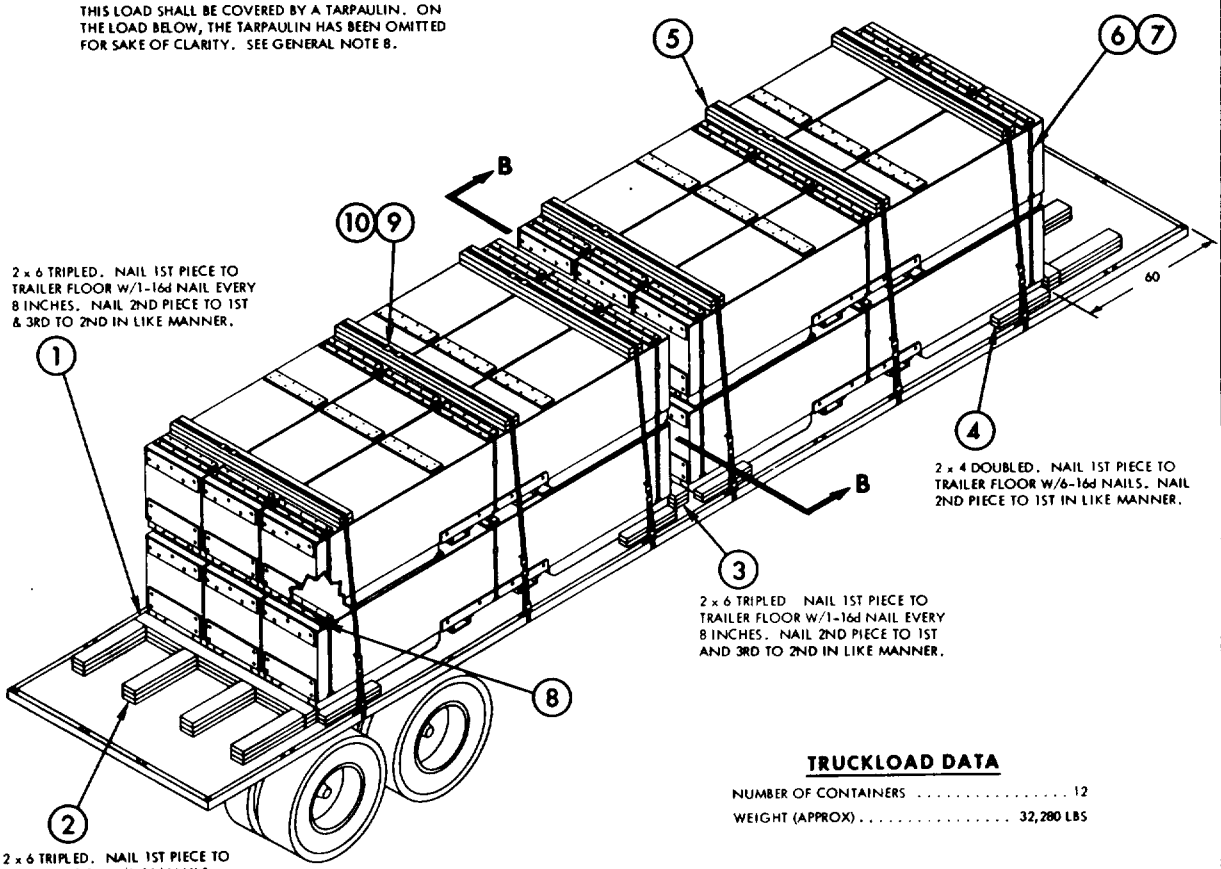
FTL 40 FT TRAILER (FLATBED)

NOTE

THIS FTL ILLUSTRATES THE USE OF 2 x .050 STRAPPING AS TIEDOWNS. CHAIN AND LOAD BINDERS ARE A AUTHORIZED ALTERNATE. SEE GENERAL NOTES 3, 4, 5, 6 AND 7.

NOTE

THIS LOAD SHALL BE COVERED BY A TARPAULIN. ON THE LOAD BELOW, THE TARPAULIN HAS BEEN OMITTED FOR SAKE OF CLARITY. SEE GENERAL NOTE 8.



2 x 6 TRIPLED. NAIL 1ST PIECE TO TRAILER FLOOR W/1-16d NAIL EVERY 8 INCHES. NAIL 2ND PIECE TO 1ST & 3RD TO 2ND IN LIKE MANNER.

2 x 4 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR W/6-16d NAILS. NAIL 2ND PIECE TO 1ST IN LIKE MANNER.

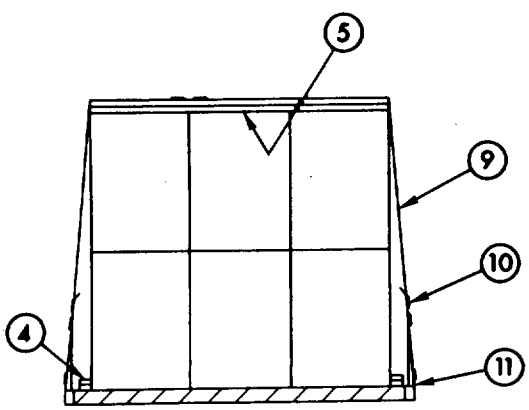
2 x 6 TRIPLED NAIL 1ST PIECE TO TRAILER FLOOR W/1-16d NAIL EVERY 8 INCHES. NAIL 2ND PIECE TO 1ST AND 3RD TO 2ND IN LIKE MANNER.

2 x 6 TRIPLED. NAIL 1ST PIECE TO TRAILER FLOOR W/6-16d NAILS. NAIL 2ND PIECE TO 1ST PIECE AND 3RD PIECE TO 2ND PIECE IN LIKE MANNER.

TRUCKLOAD DATA

NUMBER OF CONTAINERS 12

WEIGHT (APPROX) . . . . . 32,280 LBS



SECTION B-B

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER	SIZE
					NAILS	
11	STAKE POCKET PAD (STRAPPING)	2 x .050 x 24	12	-	-	-
10	SEAL	FOR 2 IN. STRAP	48	-	-	-
9	TIE-DOWN STRAP	2 x .050 x 26 FT	6	-	-	-
8	ANTI-SKID PLATE	NO. 2	48	-	-	-
7	SEAL	FOR 1 1/4 IN. STRAP	24	-	-	-
6	UNITIZING STRAP	1 1/4 x .035 x 19 FT	12	-	-	-
5	PROTECTOR BOARD	2 x 6 x 84	12	DBLD & LAMINATED	5	8d
4	SLEEPER	2 x 4 x 24	16	SEE FIELD NOTE		
3	CROSSMEMBER	2 x 6 x 84	3	SEE FIELD NOTE		
2	CLEAT	2 x 6 x 30	24	SEE FIELD NOTE		
1	END CROSSMEMBER	2 x 6 x 84	6	SEE FIELD NOTE		

LIST OF MATERIALS AND NAILING DATA

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UNITIZING PROCEDURE

1. WHEN REQUIRED BY THE FTL OR LTL REQUIREMENTS OF THIS STANDARD, THE TWO HIGH STACKED CONTAINERS SHALL BE UNITIZED AS SHOWN.
2. ANTI-SKID PLATES SHALL BE USED BETWEEN THE CONTAINERS, TWO SIGNODE NO. 2 ANTI-SKID PLATES OR EQUAL SHALL BE USED AT EACH CORNER AS SHOWN.

WARNING

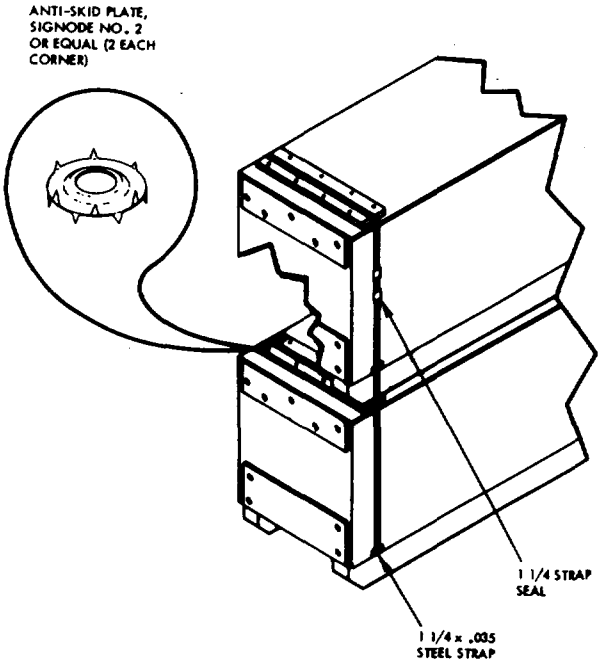
WHEN POSITIONING ANTI-SKID PLATES ON THE LOWER CONTAINER, POSITION THE ANTI-SKID PLATES SO THAT THEY DO NOT INTERFERE WITH THE STRAP USED TO RETAIN THE CONTAINER COVER.

3. THE 1 1/4 INCH STEEL STRAPS, POSITIONED AS SHOWN, ENCIRCLE THE STACKED CONTAINERS PASSING THROUGH THE STRAP SLOTS IN THE SKIDS OF THE BOTTOM CONTAINER.

NOTE

THE UNITIZING STRAPS ARE POSITIONED DIRECTLY OVER THE STRAPS HOLDING THE CONTAINER COVER IN PLACE.

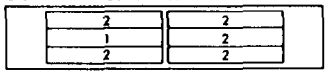
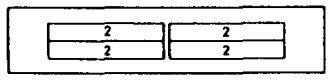
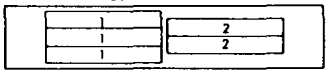
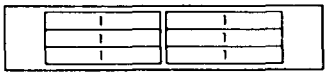
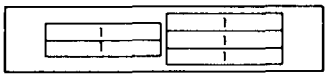
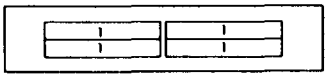
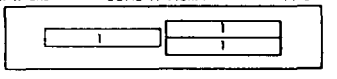
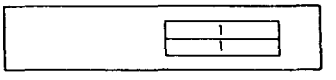
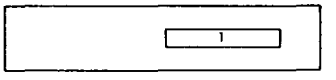
4. TENSION STRAPS AND SEAL WITH TWO DOUBLE CRIMPED SEALS.



FTL 40 FT TRAILER (FLATBED)

1. WHEN REQUIRED TO SHIP LESS-THAN-TRUCKLOAD (LTL), SELECT FROM TABLE I THE CORRECT LOAD PATTERN FOR THE NUMBER OF CONTAINERS TO BE SHIPPED AND POSITION THE CONTAINERS ON THE TRAILER ACCORDINGLY.

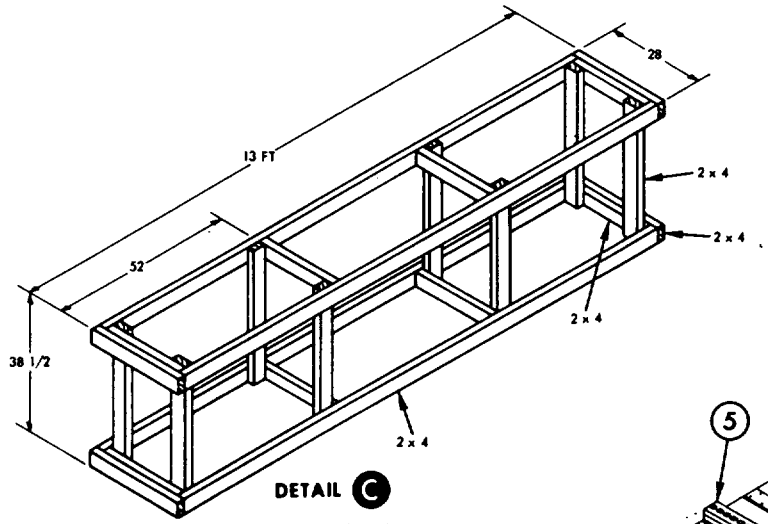
TABLE I

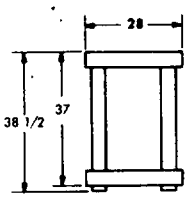
NO. OF CONTAINERS	LOAD PATTERN*	FWD
11		
10		
9		
8		
7		
6		
5		
4		
3		
2		
1		

* LOAD PATTERN SHOWS FLOOR PLAN FOR TRAILER. NUMBER INDICATES THE NUMBER OF CONTAINERS IN A LAYER.

2. BLOCK AND BRACE THE LTL USING THE PRINCIPLES SHOWN FOR THE FTL ON PAGE 3 AND THE LTL FOR TEN CONTAINERS ON PAGE 5. WHEN LOADING ELEVEN CONTAINERS USE DETAIL C TO FILL THE VOID IN THE SECOND LAYER. STRAP FILLER ASSEMBLY TO BOTTOM CONTAINER.
3. STACKS OF 6 CONTAINERS REQUIRED 3 TIE-DOWNS PER STACK. STACKS OF 5 OR LESS CONTAINERS REQUIRE 2 TIE-DOWNS PER STACK.

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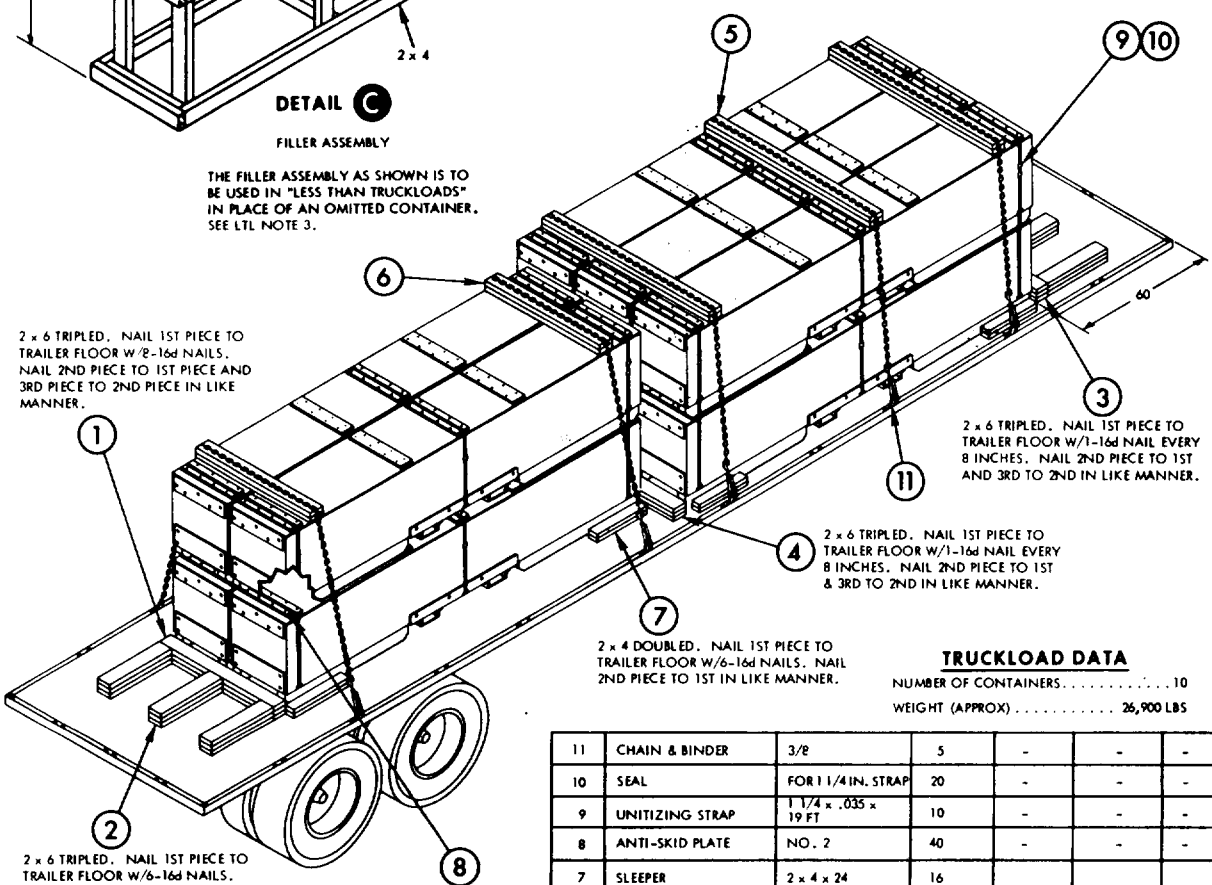


END VIEW

DETAIL C

FILLER ASSEMBLY

THE FILLER ASSEMBLY AS SHOWN IS TO BE USED IN "LESS THAN TRUCKLOADS" IN PLACE OF AN OMITTED CONTAINER. SEE LTL NOTE 3.



2 x 6 TRIPLED. NAIL 1ST PIECE TO TRAILER FLOOR W/8-16d NAILS. NAIL 2ND PIECE TO 1ST PIECE AND 3RD PIECE TO 2ND PIECE IN LIKE MANNER.

2 x 6 TRIPLED. NAIL 1ST PIECE TO TRAILER FLOOR W/1-16d NAIL EVERY 8 INCHES. NAIL 2ND PIECE TO 1ST AND 3RD TO 2ND IN LIKE MANNER.

2 x 6 TRIPLED. NAIL 1ST PIECE TO TRAILER FLOOR W/1-16d NAIL EVERY 8 INCHES. NAIL 2ND PIECE TO 1ST & 3RD TO 2ND IN LIKE MANNER.

2 x 4 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR W/6-16d NAILS. NAIL 2ND PIECE TO 1ST IN LIKE MANNER.

2 x 6 TRIPLED. NAIL 1ST PIECE TO TRAILER FLOOR W/6-16d NAILS. NAIL 2ND PIECE TO 1ST PIECE AND 3RD PIECE TO 2ND PIECE IN LIKE MANNER.

TRUCKLOAD DATA

NUMBER OF CONTAINERS 10

WEIGHT (APPROX) . . . . . 26,900 LBS

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER NAILS	SIZE NAILS
11	CHAIN & BINDER	3/8	5	-	-	-
10	SEAL	FOR 1 1/4 IN. STRAP	20	-	-	-
9	UNITIZING STRAP	1 1/4 x .035 x 19 FT	10	-	-	-
8	ANTI-SKID PLATE	NO. 2	40	-	-	-
7	SLEEPER	2 x 4 x 24	16			
6	PROTECTOR BOARD	2 x 6 x 56	4	DBLD & LAMINATED	4	8d
5	PROTECTOR BOARD	2 x 6 x 84	6	DBLD & LAMINATED	5	8d
4	CROSSMEMBER	2 x 6 x 84	3	SEE FIELD NOTE		
3	FWD CROSSMEMBER	2 x 6 x 84	3	SEE FIELD NOTE		
2	CLEAT	2 x 6 x 30	21	SEE FIELD NOTE		
1	AFT CROSSMEMBER	2 x 6 x 56	3	SEE FIELD NOTE		

LIST OF MATERIALS AND NAILING DATA

NOTE

THIS LOAD SHALL BE COVERED WITH A TARPAULIN. ON THE LOAD ABOVE, THE TARPAULIN HAS BEEN OMITTED FOR SAKE OF CLARITY. SEE GENERAL NOTE 8.

NOTE

THIS LTL ILLUSTRATES THE USE OF CHAINS AND LOAD BINDERS. 2 x .050 STRAPS ARE AN AUTHORIZED ALTERNATE. SEE GENERAL NOTES 3 AND 4.

LTL 40 FT TRAILER (FLATBED)

REVIEW ACTIVITY:
NAVY - OS

PREPARING ACTIVITY:
NAVY - OS
(PROJECT NO. 8140-N292)

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POSTAGE AND FEES PAID

OFFICIAL BUSINESS

Commanding Officer
Naval Weapons Station Earle
Naval Weapons Handling Laboratory (803)
Colts Neck, New Jersey 07722

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-STD-1320-103 (NAVY)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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